

Organizing Your Training: The Long-Range Plan

Ultramarathon (’el-tr e-’ma-r e-thän\ n.): a footrace longer than a marathon

Ask any runner what an ultramarathon is, and this is the definition you’re likely to hear. While it may technically be correct, from a training perspective, an ultramarathon cannot be thought of as merely longer than a marathon. Ultramarathon runners cannot merely extend marathon training programs found in books and lay publications because the stressors and success factors in an ultramarathon are not solely greater those in a marathon. Successful preparation for an ultramarathon incorporates correct mileage, volume, and intensity—all key pieces of training in any footrace. However, unlike marathon training, ultramarathon training needs to include vertical specificity, terrain specificity, training of the gut, adaptation to environmental stressors like heat and altitude, familiarization with equipment like packs and poles, and other considerations not routinely considered in marathon training.

Despite all of these unique needs, organizing your season does not start with the demands of the event, a specific type of interval, or any other physiological phenomenon. It is actually quite the opposite. Yes, training over the long term and how you organize it are important, and I promise I will get to that. In fact, since the launch of the first edition of this book, this chapter has proven to be the most popular, talked about, and discussed. When I go to speaking events, more questions, comments, and discussion revolve around when to do what workout than anything else. However, before you start that first workout, take some time to organize the nontraining picture. I trust you will put in the work—so before that, put the pieces in place that will maximize the results of the physical work you will do down the road (or trail, as it might be).

Choosing Your Event

Every winter, thousands of ultrarunners eagerly await the results of the coming summer’s race lotteries. Some athletes take a “throw your hat in all the rings” approach, entering every conceivable lottery option and letting fate sort it out. Others hedge their bets on one lottery, trying to stack the odds in their favor for a singular desired outcome. Whatever your approach, organizing your training should always start with one question: “What events deeply resonate with me?”

The answer to that question has many facets. You can’t control the random outcome of the major ultra lottery processes, but you can control which events you focus on. Ultrarunners these days have their pick of different racing options. Some events have elaborate race management and support systems; others are more low-key. Some are “fast”; others are “slow.” Some are in high alpine environments; others are in the desert. Whatever your preferences regarding an event’s management style, terrain, and environment, chances are you can find a race that suits your desires. Organizing your training starts with choosing the events you have a deeper connection with, because those will enhance your emotional engagement with the training process. The event might attract you because of its difficulty. Maybe you have a history in the region or an attraction to its flora and fauna. Whatever the case, take the time to ensure that the race is meaningful to you on an emotional level, not merely a box to check.

After you have determined what events are going to rile you up, it’s time to bring the people you live and run with into the loop. Rarely does one run an ultra without support from other people. Your family, friends, colleagues,

and running groups can all enhance the outcome of the events you have chosen. You may even depend on these people to crew and pace you come race day. So, get some firepower in your corner. Tell your family, friends, and fellow runners what you are training for and what your goals are. If your colleagues know what the heck you are doing with your free time, it will make sense when you walk into the office still wearing your headlamp from your morning run. Your chosen peer group can help you out, even if they don't run a step with you in training.

Strategies for Long-Range Planning

When I was in college, we had the following weekly routine:

- MONDAY: rest day
- TUESDAY: 5 x 1000 meter repeats around a grass field (similar to RunningIntervals)
- WEDNESDAY: 75-minute EnduranceRun
- THURSDAY: 6-mile TempoRun
- FRIDAY: 60-minute RecoveryRun
- SATURDAY: 8K cross-country race or 5K race on the track
- SUNDAY: 90-minute EnduranceRun

Without fail, our entire team had this routine on repeat for nine months during the cross-country, indoor, and outdoor track seasons. Sure, every once in a while we would add some trivial wrinkle, like substituting an eight-mile TempoRun for the six-mile TempoRun, or 400 m repeats instead of 1000s. But the general weekly architecture remained the same, keying in on one RunningInterval workout, one TempoRun, a race (where the intensity would resemble the RunningIntervals), and a long run. This is commonly referred to as a “mixed-intensity micro-periodization cycle,” which is an overcomplicated way of saying that your key workouts during the week are at a variety of intensities (RunningIntervals and TempoRun, in this case). Just about every collegiate cross-country program uses some small variant of this training strategy. It works well for college-aged athletes, partially because they are early in their training arc and partially because their twenty-year-old physiology will adapt to just about anything. That mixed-intensity training structure has been passed down post-collegiately for generations and made its way into the marathon and even ultramarathon realms, albeit with small caveats and nuances that claim to be markedly superior (like substituting 800 m repeats for 1000s or using a progressive TempoRun as opposed to a standard TempoRun).

While this type of training structure is almost universal in the running world, it is far less common in other endurance sports like cycling and triathlon. I have never understood why. Cyclists, runners, and triathletes all

prosper under the same type of endurance phenotype: a robust cardiovascular engine. A strong cardiovascular engine that can process copious amounts of oxygen and deliver it to working muscles is a universal requirement for all endurance sports, even if the modalities are different. Logically, the training structure means to the cardiovascular end should be equally universal. Yet, this has historically not been the case. While many runners have used the mixed-intensity architecture I experienced in college, cyclists over the last twenty years have leaned more on a “block” structure, focusing each month or training phase on one type of intensity (TempoRun, for example). Triathletes took a different take entirely, using a hybrid of both to balance development across the three disciplines.



FIGURE 10.1 Example of mixed-intensity periodization: a way of organizing training where an athlete does workouts at a few or several different intensities during the week.



FIGURE 10.2 Example of block intensity periodization plan: a way of organizing training where an athlete does workouts at similar intensities during the week.

So, which of these strategies will give you more bang for your training buck? Advocates for a mixed-intensity periodization commonly boil it down to a “use it or lose it” philosophy—meaning if you don’t stress your physiology in any one specific area (i.e., RunningIntervals that predominately stress VO2 max) for as little as a few weeks, you will lose your hard-earned fitness across that specific piece of your physiological spectrum. Advocates for block-style architecture point to the fact that because you are concentrating the physiologic stress in one area, the adaptation is likely to be greater. Research slightly favors block-style architecture in both laboratory and field settings for sports such as cycling, kayaking, and cross-country skiing (García-Pallarés 2010; Mølmen et al. 2019) and has a biological basis of support (Goutianos 2016; Issurin 2018), although the extrapolation for ultramarathon runners remains a source of debate.

For athletes I work with, and for the remainder of this book, I will generally favor a block-style training structure. Experientially I have found block-style training superior for well-trained, experienced athletes, which nearly every ultrarunner is. In addition, most ultramarathon races are performed in a very narrow range of intensities, and therefore it behooves an athlete to maximize development across that specific range, even if it is at the expense of some other area of physiology. In other words, you should be OK with the “lose it” criticism levied at the mixed-intensity philosophy because what you are losing is not deterministic in ultrarunning, so long as your long-range plan is structured correctly.

Creating a Long-Range Plan

Open any book on endurance coaching and training and you are sure to see three words: “foundation,” “preparation,” and “specialization.” The history behind these words goes back many decades to some of the founding fathers of periodization who used these terms to compartmentalize training into discrete sections. These books describe the foundation phase as being reserved for building an aerobic base, similar to laying the foundation for a house. The preparation phase “develops your cardiovascular fitness, readying you for higher-level work.” Finally, the specialization phase is reserved for higher intensities that are “specialized” to the event.

I dislike these generalizations and get enormously frustrated with the labels, which don’t accurately describe the process. First off, building an athlete is not like building a house, where certain things have to be done in a very specific order. Secondly, isn’t an athlete always “preparing?” I would rather name each training phase based on what it is—specifically by referring to the particular aspect of physiology the athlete is focusing on the most: endurance (EnduranceRuns and SteadyStateRuns), lactate threshold (TempoRuns), or VO2 max (RunningIntervals). Make no mistake, you are always training all aspects of your physiology. However, for the purpose of organizing a season, I prefer to define the segments of the year by the workouts you are doing and, thus, the primary physiological adaptation you’re after during each period of time.

STARTING AT THE END

When I design training plans for athletes, it is always a “pick and choose” exercise based on what is going to benefit the athlete the most. I have an unlimited number of workouts to choose from: speed workouts, long runs, fartleks, ladders, progressive tempos, strength training, cross-training, plyometrics, downhill intervals, uphill intervals, and more. However, after the planning process is all said and done, I end up using only a small handful of the arrows in my quiver, defined primarily by the five workouts described in Chapter 9. You always have far more options for how to improve than you have practical ones. These five primary workouts set the framework of how to organize a season.

Once athletes pick their goal events and determine the time frame they have available for preparation, I begin the process of designing a long-range plan (LRP). These plans offer a season-long snapshot of what I want an athlete to focus on at any point during the season. The plans are generalized in the sense that their level of detail is limited to weeks (not days), overarching training intensities (RunningIntervals, TempoRun, and SteadyStateRun), and other miscellaneous components of training such as the amount of vertical, type of surface, or environmental adaptations like heat or altitude. At this stage I do not outline the specific days per week or the specific combination of intervals I want the athlete to do; that is reserved for the short-range plan, explained in the next chapter. The purpose of the LRP is to ensure that an athlete employs proper strategies throughout the year. It keeps a check on when to do peak volume and how much, when to incorporate intensity, and how the race-and-recover cycles fit into the year.

Like many aspects of training, there is no one-size-fits-all approach. Each LRP I develop is for an individual athlete based on her goals. Despite this individual nature, I rely on three common principles (in order of priority) for putting together the LRP for each of my athletes.

PRINCIPLE 1:

Develop things that are the most specific to the event closest to that event, and develop the things least specific furthest away.

Figuring out the demands of an event is a very large part of what I do as a coach. The analysis starts from the very high-level cardiovascular physiology that is specific to the event and extends all the way down to the surface characteristics of the trails or roads, the distance between aid stations, the environmental conditions, and how one copes with the stress of eating and drinking. Even though the practical applications might be overly broad or extremely narrow, the principle is the same: if an aspect is highly specific to the event, develop it closer to the event.

A classic example of this can be seen in connection with the Hardrock 100. Hardrock is very specific in that hardly anyone actually runs the uphill portions of the race. Even the fastest “runners” hike the majority of the uphill and then run the descents. Therefore, applying Principle 1, I have my athletes do the majority of their hiking training in the several weeks leading up to the event because it is so specific to the race in question. You can extend this thinking to many other areas such as workout intensity, surface, environmental conditions, and the average grades of the course. Most runners who go out and recon a course in the weeks leading up to an event are applying this principle correctly. They are training specifically for the grades, surface, and environmental conditions they will face on race day.

In contrast, I have my athletes do the least specific aspects of training as far away from the event as they can logistically manage. Again, using Hardrock as an example, I cannot think of anything less specific, or inconsequential, in that race than a runner’s velocity at VO2 max. In other words, a Hardrockers’ pace at VO2 max is likely the least important factor in the whole scheme of success. But it is still important to develop (as you’ll see in Principle 2). Being the least important should not be confused with being unimportant! This is why, from a practical standpoint, many of my Hardrockers do flat, fast, high-intensity running very early in the training season.

PRINCIPLE 2:

At some point during the season, incorporate each of these three critical workouts: SteadyStateRun, TempoRun, and RunningIntervals.

It is important to visit all the various intensities, because over the course of months athletes will reach a point of diminishing returns for any one adaptation. The various intensities build on each other throughout the year, allowing the athlete to achieve better fitness. I once had a colleague ask me why it was important to train at various intensities if the event in question had a very low intensity—say, a thirty-hour Leadville finish. To him it made all the sense in the world to train at that very low but specific intensity for as long as possible during the entire course of training. As I’ve discussed previously in this book, the flaw in this logic is that all the energy systems are connected, and there are improvements to be made at the relatively easy aerobic pace that can be achieved only by raising an athlete’s maximum sustainable intensity (lactate threshold) and maximum aerobic capacity (VO2 max). To illustrate this concept, I pointed out some of the lab data we have collected from elite and amateur ultrarunners throughout the years. One of the common themes we find in ultrarunners we test just as we start coaching them is that their sustainable intensity is very, very close to their maximum intensity. In other words, their lactate threshold intensity is nearly identical to their VO2 max intensity. Sometimes this difference is as little as 3 percent.

Practically speaking, this means that only a handful of seconds per mile separates the pace they can maintain for long periods of time (hours) from the pace they can handle for only a few minutes. Normally, as a coach, I would do a backflip if I got this result from my athletes; I’d love it if their sustainable intensity were as close to their maximum intensity as possible. That is a much-coveted outcome of the training process and, if you remember earlier in this book, running speed is a product of the fraction of VO2 max you can sustain and the cost of running ($\text{Velocity} = F \times \text{VO2 max} / \text{Cr}$). It would be a huge win from a training perspective if you could sustain 80 to 90 percent of your VO2 max like some elite marathoners can for up to two hours. However, where do you go from there? Certainly your sustainable pace cannot exceed your maximum pace.

When we see this situation in our physiology lab, the training protocol becomes quite simple: the athlete needs to first raise VO2 max and then work on lactate threshold and other lower intensities. This will give the athlete a little room between VO2 max and other intensities before they go back and work on other aspects of physiology. Targeting different intensities throughout the year allows you to make measurable gains in each area, and each time you get stronger in one area, you also gain the tools to make further improvements in the other areas.

PRINCIPLE 3:

Work strengths closer to the race and weaknesses further away.

Most runners are aware of their natural strengths and weaknesses. We come to these realizations during group runs and races; anytime you run with a companion, you get an immediate barometer. If you are passing people on the climbs and getting dropped on the technical descents, then your strength is climbing and your weakness is technical running. Therefore, according to Principle 3, you should try to improve your technical running as far away from your key event as possible and your climbing as close to the event as possible. This principle is far more applicable for elite athletes who will generally win races that are closely aligned with their strengths. For other runners, the focus should be on improving your weaknesses as far out from the race as possible.

A CLOSER LOOK AT THE LONG-RANGE PLAN

The template for the LRP is a simple color-coded spreadsheet that delineates the specific types of workouts (RunningIntervals, TempoRun, and SteadyStateRun), highervolume and lower-volume periods, critical events throughout the season, and other variables of training like amount or rate of vertical gain and loss, trail vs road, etc. I have supplied an example of this for an athlete training for the Western States 100 and also provided a blank LRP template in the Appendix.

The LRP offers me a season-long picture of what the athlete needs to work on at any given point. It also gives the athlete a heads-up on when to expect higher volume and higher intensity and when to focus on other aspects of the event such as nutrition planning. However only so much detail can go into the plan at this point. You will always need to adapt as the season goes along. Nevertheless, starting out with a solid plan from the get-go will ensure that you're keeping your priorities straight during the year, so when push comes to shove, you know what to push and what to shove!

MONTH	January				February				March				April				May				June													
RACE NAME					Rocky Raccoon 100								Lake Sonoma 50					Silver State 50					Western States Training Camp					Western States 100						
RACE PRIORITY						A									B						B												A	
PHASE GOAL						Short round of RIs					Fitness build, 4 x 15 T					SSR and EnduranceRuns																		
RECOVERY																																		
ENDURANCE																																		
STEADYSTATERUN																																		
TEMPORUN																																		
RUNNINGINTERVALS																																		
TAPER																																		
NOTES											Develop initial nutrition strategies				Pick training back up quickly after Lake Sonoma				Use Silver State as a training race				Highest volume, lowest intensity				Race nutrition strategies during all long runs				Sauna protocol			

FIGURE 10.3 An example of a Long-Range Plan.

When you are developing your LRP, think broad brushstrokes, not minutia. Find the things that are most important for you to be successful, and develop those aspects in the several weeks leading up to the race.

Applying the Three Principles of Long-Range Planning

PRINCIPLE 1:

Develop the physiology that is most specific to the event closest to the event, and develop the least specific physiology furthest away.

The Western States 100 is notorious for being a long, hot day in the mountains. Long, as in it typically takes even the best athletes over fifteen hours to finish. Hot, as in temperatures commonly exceed 100°F in the middle part of the course. And mountainous because of the 400-plus feet of elevation change per mile. Therefore, the most specific pieces of physiology this athlete needed to develop were the ability to run at a relatively low intensity

(EnduranceRun for her), to withstand the heat, and to be able to cope with the climbing and descending on the course.

In contrast, VO2 max-type RunningIntervals represent the least specific intensity for the Western States 100. There was very little chance any athlete would be running anywhere close to this intensity during Western States. Therefore, this extremely high, unsustainable intensity was “least important” (not unimportant).

Application of Principle 1

Perform SteadyStateRuns and EnduranceRuns closest to the race and RunningIntervals furthest away. Focus on heat acclimation and specific nutrition strategies closest to the race.

While this athlete did not have the opportunity to train regularly on mountainous terrain like she would encounter during Western States, she seized opportunities to overcome this by going to the Western States training camp, using the Silver State 50 as a training race, and using the 100- to 200-foot climbs in her local area as much as possible. This confluence of targeting the right intensity, heat acclimation, and training on terrain as close to the race environment as possible during the last eight weeks of training ensured she was optimally prepared for the conditions of the race.

PRINCIPLE 2:

At some point during the season, incorporate each of the three critical workouts (SteadyStateRun, TempoRun, and RunningIntervals).

The athlete’s early-season RunningInterval work and late-season EnduranceRun/ SteadyStateRun work bookended the process so, naturally, the middle could easily get filled in with TempoRun. In many five- to seven-month LRPs, a good way to facilitate the flow of training from one phase to another is to figure out the bookends first, then fill in the middle with whatever is left.

WHAT INTENSITY DO I RACE AT?

IT IS MOST COMMON FOR ULTRARUNNING race intensities to be SteadyStateRun/ EnduranceRun, with VO2 max intensity at the other—least-common—end of the spectrum. This holds true for even “shorter” ultramarathons lasting four to six hours. Exceptions are rare and consist of skyrunning races as well as 50K and 50-mile events for elite runners. Elites in those race distances run for a shorter duration (because they are faster) and can run at higher intensities for longer periods. In these races, the elites can actually run within 10 to 15 percent of their threshold intensity for much of the race.

Application of Principle 2

The bookends of this athlete's physiological demands mirror her strengths and weaknesses (see Principle 3). Thus, accomplishing all three principles was an easy task, as it was a function of determining what to do last, determining what to do first, and then filling in the middle.

PRINCIPLE 3:

Work strengths closer to the race and weaknesses further away.

I knew from the onset that one of this athlete's strengths was her physiology at EnduranceRun intensity. I could tell this from her performance at 100-mile competitions compared to marathon and 50K distances, as well as data from her day-to-day training. She can tolerate high training volume very well and can seemingly run forever at lower intensities. By contrast, she is not as good at races like the 5K and 10K (which rely more on VO2 max intensity), and we typically see fatigue set in quickly when she is prescribed a phase of RunningIntervals. All of this points to her strengths being more rooted in lower intensities and weaknesses rooted in higher intensities.

Application of Principle 3

Since the athlete's strengths and weaknesses lined up with the most and least important aspects of her physiology, the flow from RunningIntervals to TempoRun to SteadyStateRun was a natural one. This is not always the case. Sometimes, the time frame or other mismatch dictates that you have to pick and choose among the three principles, even leaving one of them out. More often, you skimp on one to develop another. These are judgment calls, and thus there are few right and wrong approaches. In general, though, I tend to favor training for the demands of the event as an ever-so-slightly higher priority than the other two principles.

Training Phases

Generally speaking, each training phase should last approximately eight weeks. This athlete's ultimate LRP conveniently fits into this time frame, with the exception of the early season RunningInterval phase. That phase was intentionally shortened for reasons I will describe below. If you are designing your own LRP, I encourage you to go through the following process:

Begin planning by identifying what training phase will be last. This will be the intensity that is most specific to your race and/or your strength(s). Make this final bit of training two phases of four weeks each.

Next, plan your first training phase with what is least important and/or the weakness you want to work on. Make this two phases of four weeks each.

Fill in the middle with the intensity that is left over.

Adjust your time frame for each phase based on the aspects I will soon describe.

Add in any nuance for nutrition training, heat/altitude acclimation, or any training camps.

nutrition

In reality my athletes are switching the intensity of their training every four to twelve weeks, with the number of weeks based on three distinct facets: intensity, significance, and rate of adaptation.

INTENSITY OF THE PHASE

Typically, higher-intensity phases can be shorter than lower-intensity phases. I commonly have athletes go through a VO₂ max phase that is as short as two or three weeks. It is equally common for an endurance phase that includes SteadyStateRuns and EnduranceRuns to last twelve weeks, with rest at appropriate times. This is because your physiology at high intensities adapts over a shorter time course and vice versa. Stephen Seiler (2006) has developed a useful theoretical curve to visually explain this phenomenon and its impact on performance (Figure 10.4). It neatly describes how an athlete's VO₂ max and lactate threshold can improve and plateau over time.

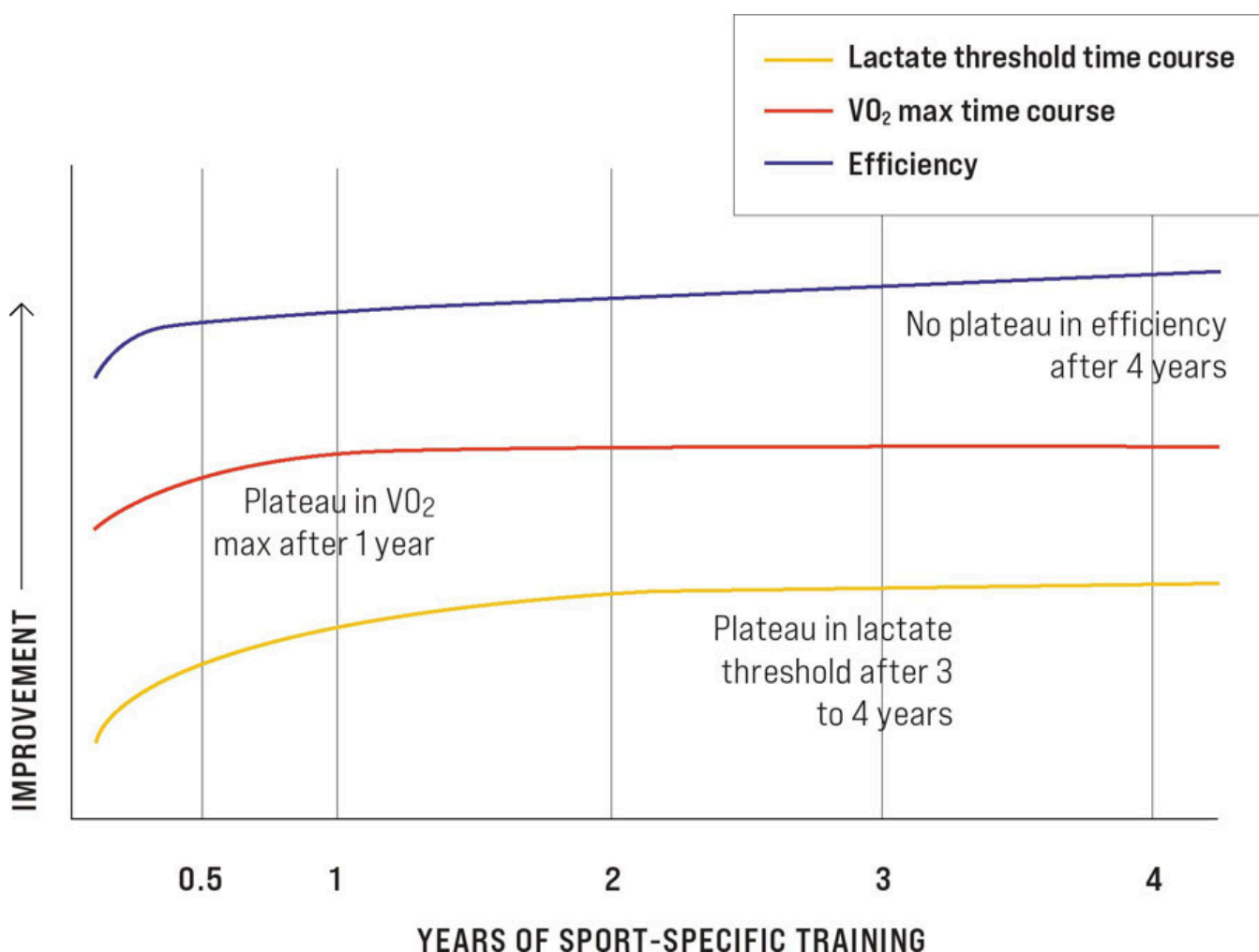


FIGURE 10.4 Time course for training adaptation. Adapted from Seiler 2006.

SIGNIFICANCE OF THE PHASE IN THE OVERALL PLAN

Significance is quite easy to determine and correlates to Principle 1 from designing the LRP. Quite simply, if the training aspect is more critical to success, I will have an athlete develop it closer to the race and also for a longer period (typically eight weeks or more). If the aspect is less important, I have the athlete work for a shorter period (eight weeks or less) and typically further away from the event.

POINT OF DIMINISHING RETURNS

The concept of diminishing returns can be illustrated by the example of eating cookies. The first cookie is delicious, melting in your mouth with a warm, buttery, sugary goodness. By the time you eat the fourth or fifth cookie, meh, it's OK but not nearly as satisfying as the first. The enjoyment you get from the treat diminishes as each subsequent cookie is devoured. Physiologically speaking, the concept of diminishing returns applies to training: the returns on your hard work diminish as time and training move forward. Additionally, as you become more fit, it takes a longer amount of time to improve less (again, see Figure 10.4). This is generally a good thing, as it means you are becoming stronger and faster. Depending on the type of stress, this point of diminishing returns can come after weeks, months, or even years. Figure 10.4 also illustrates that even though these physiological phenomena can reach their points of diminishing returns, the athlete's efficiency (and therefore performance) can still improve.

While the figure specifically describes physiological changes at lactate threshold and VO₂ max as we would measure them in a lab, the basic idea is that higher intensities produce changes in an endurance athlete more quickly but to a lesser extent, and they reach the point of diminishing returns sooner as compared with lower intensities. By knowing this simple physiological phenomenon, you can better gauge when you've reached this point of diminishing returns, which is the third facet in determining how long to focus on what stress.

BALANCING SIGNIFICANCE AND DIMINISHING RETURNS

While the general intensity of the phase can inform how long or short a phase should be, the aspects of significance and diminishing returns need to be balanced when actually constructing the details of the phase. During the planning process, significance takes priority. How important is the phase to the end result? This dominates the phase duration philosophy because the reality of training has yet to take place. You as an athlete, and I as a coach, have yet to see how the training and adaptation actually unfold. So, we make an educated guess about when our neat little blocks are going to begin and end when we formulate the LRP. Sometimes those nice, neat blocks run true to form, but sometimes they change.

Once reality takes hold we can prioritize training based on what we observe about the aspect of diminishing returns. As a coach, I evaluate training on a daily basis, looking at how fresh and fatigued an athlete is. Even more important, I use that information to determine whether an athlete is better after a rest phase, and I either keep the original LRP or alter it accordingly. It's a tricky balance, because I want to see the athlete fatigue during the course of training and then improve just after coming out of a recovery phase. Athletes need to literally get worse before they rest and thus get better; they need that amount of stress in order to adapt. Yet needing an athlete to get worse before getting better can only be taken so far before the stress is too great.

When I see an athlete fail to improve shortly after a recovery phase, it's one indication that they are ready for a different stimulus. For athletes who are designing their own training programs, I encourage you to switch in and out of training phases more quickly, rather than forcibly wringing out every last ounce of improvement. This errs on the side of caution, keeping the training stimulus novel and helping to prevent burnout. For the athletes I coach, I look at a range of variables to indicate that a change is needed and appropriate, including interval metrics, cumulative training load, acute training load, training stress scores, qualitative feedback, and workout comments from the athletes' training logs. I have outlined some of these variables in Chapter 6.

An Ultrarunner's Hierarchy of Needs

My alma matter's code of conduct was "Aggies do not lie, steal, or cheat, nor do they tolerate those that do." As we Texas A&M Aggies are a cheeky lot, we would routinely add an addendum to the code that stated: "They just exaggerate, collaborate, and borrow." Little did I know that the "collaborate and borrow" tongue-in-cheek piece would actually come in handy as I became a professional coach.

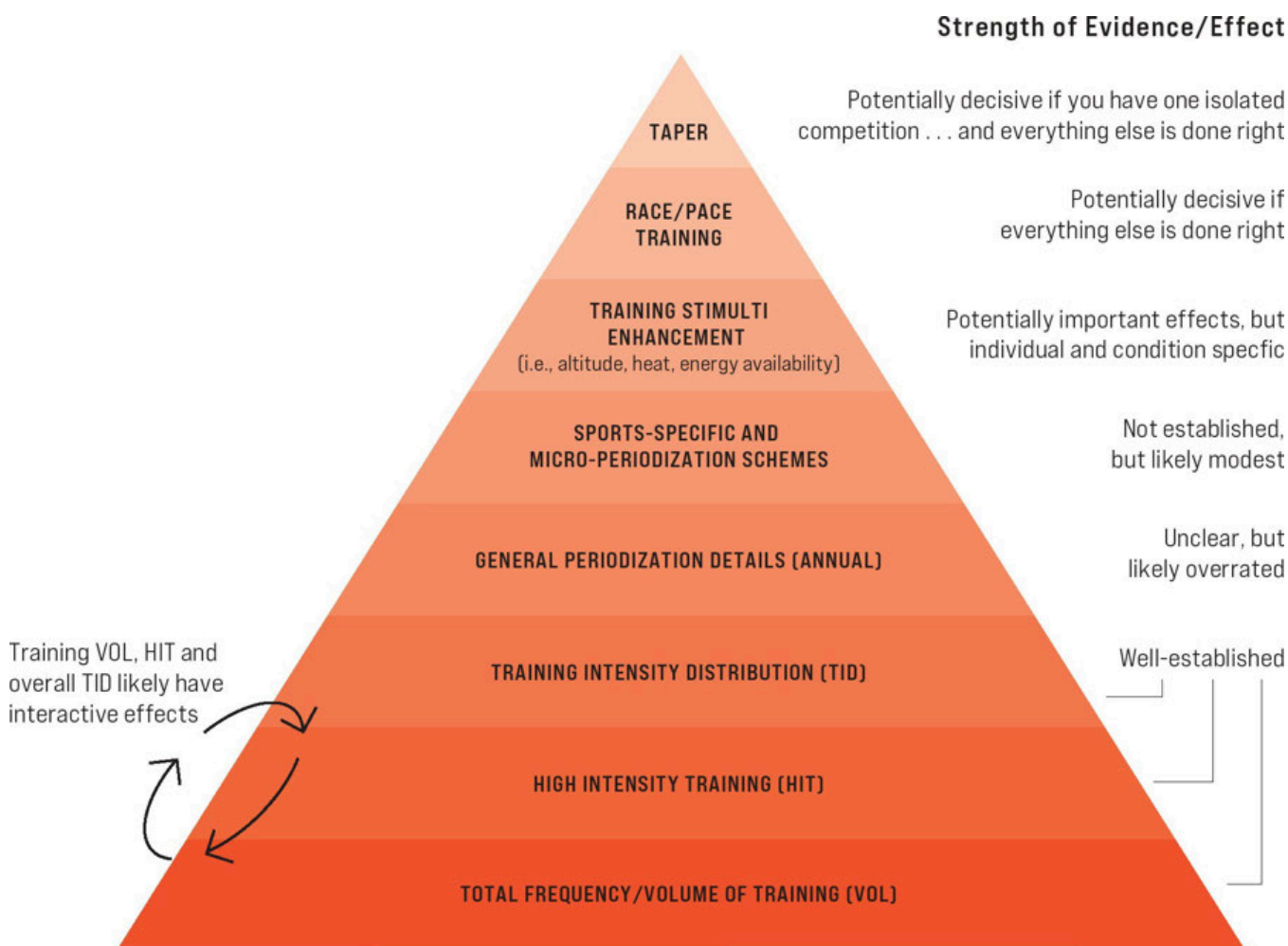


FIGURE 10.5 Hierarchy of Endurance Training Needs. Source: Seiler and Sjursen 2004.

In a 1943 paper, Abraham Maslow described human beings' basic needs and arranged them in a hierarchy, proposing that some basic needs (food, water, security) have to be met before "higher" needs (social relationships, ego, self-esteem) are addressed. Maslow's hierarchy of needs has been taught in psychology classes and re-envisioned for other purposes for decades. As a student of sports science, I was enthralled by Dr. Stephen Seiler's application of this concept to physical training. He refers to the pyramid below as the "hierarchy of training priorities for endurance athletes."

The basic premise of Seiler's hierarchy is that athletes need to accumulate a lot of training time—and lots of time at intensity below and around lactate threshold—before adding more specialized techniques. I completely agree and have preached the notion of "first things first" for years: go after the big 20-percent performance gains from hours of fundamental endurance training before you try to squeeze out a 1 to 2 percent improvement from "marginal gains." Don't add altitude training until you've done the work to develop your aerobic engine first.

As a coach, I felt the need to “collaborate and borrow” Dr. Seiler’s hierarchy a step further. While his hierarchy describes the priorities that must be addressed to increase athletic performance, I think there is a separate hierarchy that describes the priorities needed for ultramarathon athletes, and coaches working with them, to reach their full potential. As you work through your LRP, keep the following “hierarchy of ultramarathon training needs” in mind. Focus on the things that matter more. Training concepts near the big base of the pyramid will matter much more than the nuance of whether you spend eight or ten days in the sauna. The hierarchy of ultramarathon training needs is also a good reminder to include the “other” things in training that are often forgotten. Aspects like building your team and training the gut are important but often get lost in the desire to design the number of miles, amount of vertical, and timing of intensity. Do not neglect these; include them in the notes or an additional section of your LRP.

Koop’s Hierarchy of Ultramarathon Training Needs

Athletes are not science experiments, and they are not machines. Human physiology can be predicted and measured using various formulas, equations, and tests, but human performance cannot. There’s no equation for mental toughness. There’s no formula for learning to adapt when Plan A, Plan B, and Plan C all go out the window. I’ve always said coaching is a combination of art and science, and when I view the concept of a hierarchy of needs through an ultramarathon coaching lens, here’s what I come up with:

TOTAL VOLUME OF WORKLOAD

As with Seiler’s hierarchy, the first thing athletes have to do is put in the time. If you ask me what you should do to improve your performance, I’ll almost always say “train more over long periods of time.” The “over long periods of time” piece is important, because it de-emphasizes the single long run or single-week volume in favor of annual volume, which is far more impactful and mitigates injury. From a coaching perspective, this means helping athletes find or create more training volume throughout the year. I try to increase the number of sessions per week or overall number per month. For time-crunched athletes who can’t increase total weekly hours or the number of training sessions per week, I try to make some runs longer. Longer runs are helpful for generating a bigger stimulus, and for people with tight schedules I sometimes reduce the number of runs per week by consolidating two shorter training sessions into one longer run. They get the same number of hours per week, but rearranged to add duration to an individual session.

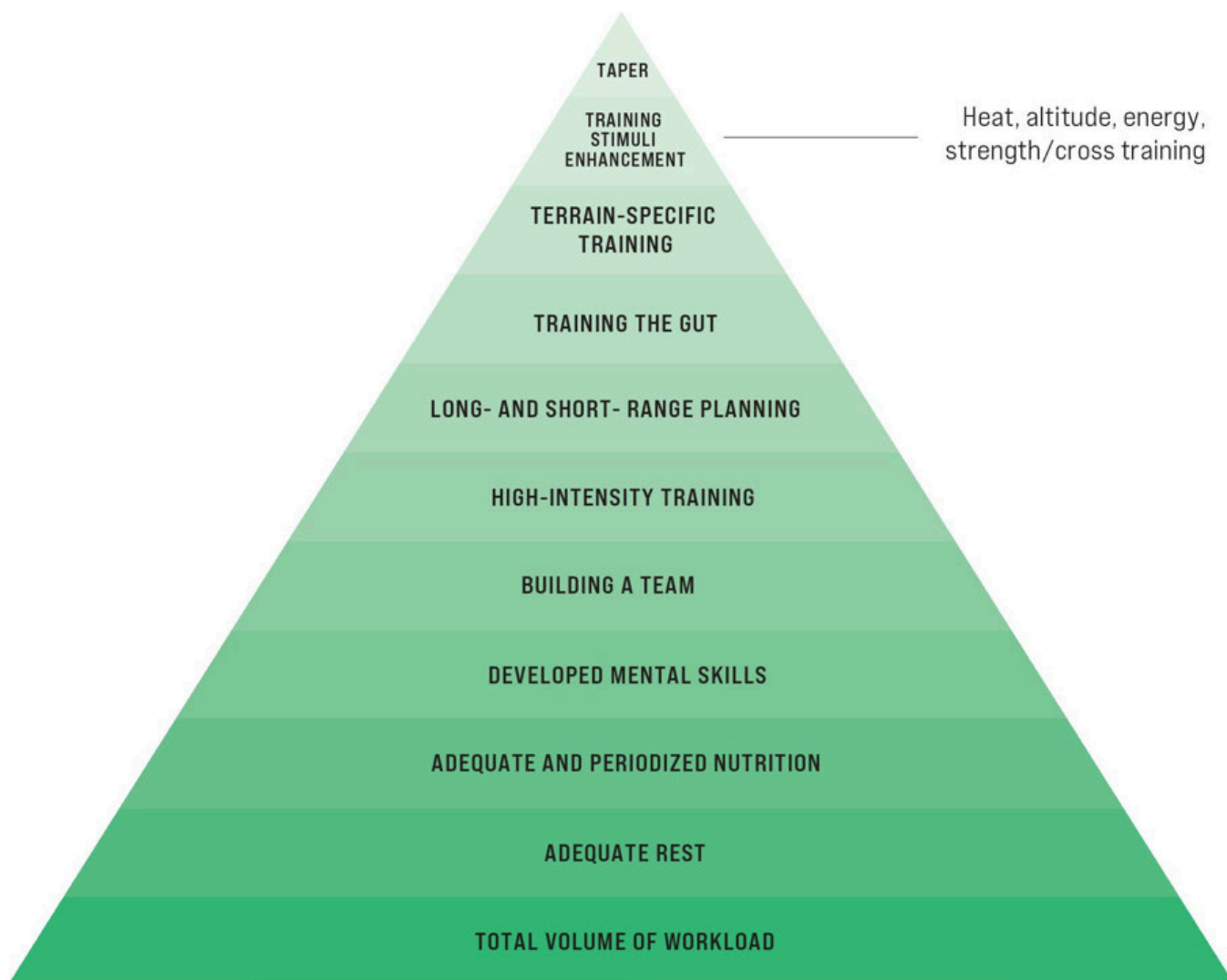


FIGURE 10.6 Koop's hierarchy of ultramarathon training needs.

ADEQUATE REST

Rest is implicitly included in Seiler's hierarchy because it is integral to structuring training, but I break it out into a distinct coaching need because it is typically harder to get athletes to rest than it is to get them to go full gas during an interval workout. Rest is boring, and high-energy, driven athletes have trouble taking it easy. When I ask a roundtable of CTS coaches the number one thing they wish their athletes would do more, it's rest.

WHOLESOME NUTRITION, MOSTLY PLANTS, PERIODIZED A PLUS

If an athlete is putting in the time and following the plan in terms of work and recovery, they also need to make sure they're fueling to meet the demands of training. The foodstuff you ingest literally becomes the fuel for the machinery in your body that propels you forward. I don't preach any particular diet, but diets that eliminate or demonize particular food groups or macronutrients are problematic for athletes at best and harmful at worst. The most important guidelines I try to reinforce with endurance athletes are:

- Consume adequate energy. What you eat only matters once you're meeting your overall energy needs for training and recovery.
- Eat more plants and fewer animals. You don't have to be solely plant-based, but adding more plants is a good move. There's nothing wrong with animal protein, but you can get all you need from plants, too.
- Don't be afraid of carbohydrates. Consume more fresh fruit and vegetables, and save concentrated carbohydrate sources like processed grains for times when energy expenditure from training is highest.
- Periodize nutrition. Your training volume and intensity change during the year, and your caloric intake—and your food choices to a lesser extent—should change, too.

DEVELOPED MENTAL SKILLS

Your physiological characteristics determine the maximum amount of exercise capacity you ultimately have. Your psychological skillset determines how much of that total capacity you can utilize. This goes far beyond being “tough,” “gritty,” and the cliché descriptors we assign to athletes who outperform their physiologically equivalent peers. Mental skills for ultrarunning are an imperative part of the sport and are described in further detail in Chapter 15.

BUILDING A TEAM

There is only so much an athlete can accomplish on their own before they need to build a strong support system. This includes some or all of the following: family, friends, training partners, a coach, a physical therapist, a massage therapist, a nutrition professional, and a physician. Many athletes resist this step, thinking they are “not enough of an athlete” to warrant substantive support. As a coach, one of my roles is to help an athlete build a support network, actually utilize it, and recognize the value it delivers. I routinely build relationships with my athletes' families and support networks because I know it is all part of the big wheel that keeps them going. When everything is going great, it may seem unnecessary. But when everything goes sideways, athletes without an established support system get swept way off track.

HIGH-INTENSITY TRAINING AND DISTRIBUTION OF TRAINING INTENSITY

High-intensity training is an important component of training for endurance athletes— even ultra-endurance athletes—but from the time-at-intensity standpoint it actually accounts for maybe 20 percent of training sessions and only about 10 percent of total training hours per year. If you run ten hours a week (six-hundred minutes), you'd have to average sixty minutes of work above lactate threshold just to hit 10 percent. Even though I am a coach known for—and sometimes criticized for—using VO2 max interval workouts with ultramarathon runners,

the actual distribution of training intensities I prescribe is heavily weighted toward low-intensity training with strategic use of high-intensity workouts.

LONG- AND SHORT-RANGE PLANNING

Long- and short-range planning is essentially the application of variety in training in order to change the stimulus. On an annual basis this means training aspects of fitness that are less specific to your goal event farther out from the event, and then training more event-specific aspects of fitness as the event draws closer. On a phase or weekly basis, this gets down to how individual workouts are arranged on days of the week. Planning plays a role in fine-tuning an athlete's performance for events, but coaches have to be careful not to give rigid schemes more importance than they are due.

TRAIN THE GUT

Gastrointestinal distress is a leading cause of DNFs in ultramarathon events, and I view that as a consequence of inadequate preparation. The gut can and should be trained to handle the volume of food and fluids an ultramarathoner needs to consume over a six-, twelve-, or thirty-hour period. If you've done everything right in terms of training, rest, and support, you'll be able to reach the point in an ultra-endurance event where your gut may be the weakest link. If you haven't done those other things first, you won't last long enough in the event for your gut to be an issue.

TERRAIN-SPECIFIC TRAINING

2016 Western States winner Kaci Lickteig is a great example of why terrain-specific training occupies an advanced step in the hierarchy of ultramarathon needs. She trains in Nebraska, without access to big mountains, yet she's won the Western States Endurance Run and other ultras with tens of thousands of feet of climbing. Spending all your time running in the mountains doesn't automatically make you a great mountain runner, especially if your training is not addressing the fundamentals first. As a coach, my goal is to put the fittest athlete on the start line, because fitness is an athlete's most important tool for creating the ability to adapt to the specific challenges of an event. If you can match a goal event's elevation gain and climbing in training, that's awesome, but develop the biggest aerobic engine you can before worrying about terrain-specific training.

TRAINING STIMULI ENHANCEMENT

Training stimuli enhancements like altitude training and heat acclimation sit near the very tip-top of the pyramid, after everything else has been done. Does altitude training work? You bet, but only if the athlete is healthy, already fit, and well rested. Is heat acclimation important? For races in hot weather, yes. But it doesn't take that long or that much effort to achieve most of the benefits from heat acclimation, so it shouldn't displace fundamental training months before your event. The same can be said for strength training (more on that in Chapter 11). The trouble is, chasing improvements in these areas, sometimes referred to as marginal gains, is sexier than going out for another four by ten-minute TempoRun workout. And coaches fall for the appeal of marginal gains, too, often to their athletes' detriment.

TAPER

“Taper tantrums” are real and derail many athletes right before major events. Keeping an athlete on track in the final two weeks before a major race is perhaps a coach’s most important role, but not because of the potential 1 to 2 percent improvement in performance that might come from a perfect taper. Early in 2018, Dylan Bowman won the Ultra-Trail Mt. Fuji by a mere couple of minutes. The percent he won by was in this cliché 2 percent area. Did the taper make the difference? Maybe, but it is certainly not the whole picture.

SLEEP DEPRIVATION TRAINING FOR 200s?

THE EMERGENCE OF 200-MILE RACES has raised the collective eyebrows of the ultramarathon world. Events like the Triple Crown of 200s (the Bigfoot 200, Tahoe 200, and Moab 240 all in the same year) as well as the Tor des Géants have drawn the eyes of ultrarunners looking for their next great challenge. Indeed, as ultramarathons are not simply long marathons, 200 milers (and beyond) are not simply longer ultramarathons in terms of training, preparation, and race strategy. You can’t just run more and expect that to translate into success at longer and longer distances. So, while the general preparation outlined in this chapter can apply to any ultramarathon, there are some specific considerations athletes should take note of for runs of 200 miles and longer.

One of the more challenging characteristics of 200-mile races is that sleep deprivation will eventually become a factor, for nearly all runners, in the outcome. This has led to the logical notion that ultrarunners preparing for these events need to undertake some type of sleep deprivation training to better cope with that stress. Anecdotal strategies have emerged and range from training sessions performed overnight or after a short sleep to training after being awake for twenty-four hours. All of these strategies attempt to condition the body to a lack of sleep. They are all novel and worthy attempts to solve a critical performance problem for these events. However, there is very little, if any, scientific evidence indicating that any sort of sleep deprivation training has any meaningful physiological effect on an athlete’s ability to tolerate a lack of sleep. If anything, some of these training techniques will familiarize an athlete with how it feels to run overnight and operate with little sleep, but the extent of any benefit is limited to just that: the familiarization of locomoting with little sleep.

In addition to this, sleep deprivation training comes with a cost. You are interrupting and compromising normal training by intentionally introducing a sleep-deprived state. The quality of sleep-deprived sessions is compromised on the order of about 2 to 10 percent (Oliver et al. 2009; Roberts et al. 2019; Skein et al. 2011). Training needs to be altered before and after the sleep-deprived session in order to accommodate pre- and postworkout rest. Athletes also might be more vulnerable to injury and illness after training in a sleep-deprived state. All of this effort costs valuable training time and begs the question, “Is it even worth the effort?” In my opinion the balance between advantages and potential downsides of sleep deprivation training favors normal day-to-day training as a means of maximizing the athlete’s physical capacity. If anything, an overnight session or two might be warranted simply to test lighting and other equipment, as well as to have some degree of familiarization with running for prolonged periods with a headlamp.

Specific sleeping strategies for 200-milers will be discussed in Chapter 16.

The more important part of the taper is not screwing up everything the athlete worked so hard to build. Every year I see runners training on the track in Chamonix in the days before Ultra-Trail du Mont-Blanc. This is silly because it is the least event-specific activity they could do. It’s a coach’s job to prevent athletes itchy for competition from being their own worst enemies. In an ideal world, the athlete enters their taper period confident they have done all

the work necessary to be at their best, and that confidence allows for relaxation and stress reduction, all of which enables them to arrive at the start line rested and ready to turn all their preparation into action.

LAST WORD ON THE HIERARCHY OF ULTRAMARATHON TRAINING NEEDS

Athletes rarely achieve breakthrough performances purely because of their physical training. Getting to the start line with the best possible conditioning is important, but it's not everything. To realize your full potential, you have to go to the start line with a full toolbox of mental, physical, and emotional skills so you are as prepared as possible to deal with the unpredictable nature of endurance sports. Some of those tools are more or less fundamentally important for success, and some take more or less time to develop. The hierarchy of ultramarathon training needs provides a visual guide to rank ordering seemingly unrelated aspects of preparation into one hierarchy. Your best day doesn't happen because you have the highest VO2 max or because you have optimized fat oxidation; it happens because you have prepared yourself for whatever a race can throw at you, and you have prepared more for the things that matter more.

KEY POINTS FOR LONG-RANGE PLANNING

Block periodization will be superior to mixed periodization for most ultrarunners.

When you are working on your LRP, start at the end (your goal event) and work backward chronologically.

Do the things that are most specific to the race closest to the race. Do the things that are least specific to the race further away. This normally results in a plan that moves from high intensity and low volume to high volume and low intensity as the season progresses.

At some point during training, do one training phase of each critical intensity (SteadyStateRun, TempoRun, and RunningIntervals).

Work your weaknesses earlier and your strengths later.

Start with eight-week training blocks, divided in half with a recovery week every fourth week, working one intensity at a time.

Shorten a block if:

- **The intensity is high**
- **The intensity is less important**

Lengthen a block if:

- **The intensity is low**
- **The intensity is more important**

If you are designing your own training, err on the side of short training cycles and switching the intensity more frequently.

You have a hierarchy of needs as an ultrarunner. The total amount of work you do and adequate rest are the most important.

